

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. IV YEAR**  
**AUTOMOBILE ENGINEERING**

**VII SEMESTER**

**R22**

| Course Code | Title of the Course                                           | L  | T | P/D | CH | C  |
|-------------|---------------------------------------------------------------|----|---|-----|----|----|
| 22PC1AE401  | Advanced Driver Assistance Systems                            | 3  | 0 | 0   | 3  | 3  |
| 22PC1AE402  | Automotive Embedded Systems                                   | 3  | 0 | 0   | 3  | 3  |
|             | PROFESSIONAL ELECTIVE – III                                   |    |   |     |    |    |
| 22PE1AE401  | Artificial Intelligence for Autonomous and Connected Vehicles | 3  | 0 | 0   | 3  | 3  |
| 22PE1AE402  | Two and Three-Wheeler Technology                              |    |   |     |    |    |
| 22PE1AE403  | Advanced Manufacturing Technologies                           |    |   |     |    |    |
| 22PE1ME401  | Computational Fluid Dynamics                                  |    |   |     |    |    |
| 22PE1ME412  | Design for Manufacturing and Assembly                         |    |   |     |    |    |
|             | PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL               |    |   |     |    |    |
|             | Emerging Technologies Stream                                  | 3  | 0 | 0   | 3  | 3  |
|             | Design Stream                                                 |    |   |     |    |    |
|             | Automotive Stream                                             |    |   |     |    |    |
|             | Manufacturing Stream                                          |    |   |     |    |    |
|             | Thermal Stream                                                |    |   |     |    |    |
|             | OPEN ELECTIVE – III                                           | 3  | 0 | 0   | 3  | 3  |
| 22PC2AE402  | Automotive Embedded Systems Laboratory                        | 0  | 0 | 2   | 2  | 1  |
| 22PC2AE411  | Vehicle Maintenance and Testing Laboratory                    | 0  | 0 | 2   | 2  | 1  |
| 22PW4AE401  | Major Project Phase – I                                       | 0  | 0 | 6   | 6  | 3  |
| Total       |                                                               | 15 | 0 | 10  | 25 | 20 |

L – Lecture      T – Tutorial      P – Practical      D – Drawing  
C – Credits      SE – Sessional Examination      CA – Class Assessment  
SEE – Semester End Examination      D-D – Day to Day Evaluation  
CP – Course Project      PE – Practical Examination

CH – Contact Hours/Week  
ELA – Experiential Learning Assessment  
LR – Lab Record

**VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD**  
**B.TECH. IV YEAR**  
**AUTOMOBILE ENGINEERING**

**VIII SEMESTER**

**R22**

| Course Code | Title of the Course                                   | L | T | P/D | CH | C  |
|-------------|-------------------------------------------------------|---|---|-----|----|----|
|             | PROFESSIONAL ELECTIVE – V                             |   |   |     |    |    |
| 22PE1AE404  | Electric Vehicle Charging Infrastructure Technologies | 3 | 0 | 0   | 3  | 3  |
| 22PE1AE405  | Automotive Noise and Vibration Control                |   |   |     |    |    |
| 22PE1AE406  | Farm Power and Machinery                              |   |   |     |    |    |
| 22PE1AE407  | Green Hydrogen Technologies                           |   |   |     |    |    |
| 22PE1AE408  | Automotive Product Development Strategies             |   |   |     |    |    |
|             | PROFESSIONAL ELECTIVE – VI                            |   |   |     |    |    |
| 22PE1AE409  | Alternative Fuels and Pollution Control               | 3 | 0 | 0   | 3  | 3  |
| 22PE1AE410  | Automotive Biomechanics                               |   |   |     |    |    |
| 22PE1AE411  | Special Purpose Vehicles                              |   |   |     |    |    |
| 22PE1AE412  | Automotive Instrumentation and Testing                |   |   |     |    |    |
| 22PE1AE413  | Robotics in Automotive Industry                       |   |   |     |    |    |
|             | OPEN ELECTIVE – IV                                    | 3 | 0 | 0   | 3  | 3  |
| 22PW4AE402  | Major Project Phase – II                              | 0 | 0 | 22  | 22 | 11 |
| Total       |                                                       | 9 | 0 | 22  | 31 | 20 |

L – Lecture      T – Tutorial      P – Practical      D – Drawing  
C – Credits      SE – Sessional Examination      CA – Class Assessment  
SEE – Semester End Examination      D-D – Day to Day Evaluation  
CP – Course Project      PE – Practical Examination

CH – Contact Hours/Week  
ELA – Experiential Learning Assessment  
LR – Lab Record

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PC1AE401) ADVANCED DRIVER ASSISTANCE SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis, Automotive Engines, Automotive Electrical and Electronics

#### COURSE OBJECTIVES:

- To know the significance of sensors and cameras used in ADAS
- To understand various telematics and comfort systems
- To recognize the importance of safety systems in automobiles
- To understand the role of ADAS in autonomous vehicles
- To acquire knowledge on testing and validation methods of ADAS systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Describe various sensors and cameras and their integration into ADAS

**CO-2:** Explain different types of telematics and comfort systems

**CO-3:** Illustrate different types of safety systems

**CO-4:** Explore the role of ADAS in autonomous vehicles

**CO-5:** Explain the testing and validation methods of ADAS systems

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-2 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-3 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-4 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-5 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |

#### UNIT-I:

**Sensors in ADAS:** Role and importance of ADAS, sensors-types, working principles, comparison. Sensor fusion - techniques, challenges and its application in ADAS. Vision based ADAS systems- cameras - front, rear, side-view, surrounded-view. Image processing for ADAS.

#### UNIT-II:

**Telematics and comfort systems:** Global positioning system, geographical information systems, navigation system, remote access, emergency calling systems, vehicle maintenance and notification systems, driver health monitoring. Automatic adjustments-seat, steering wheel, rear and side view mirrors, smart cards, GSM and short-range communication systems.

**UNIT-III:**

**Safety systems:** Active and passive safety, adaptive cruise control, Co-operative adaptive cruise control, lane keep assistance, lane departure warning, collision warning systems, anti-lock braking systems, traction control systems, traffic sign recognition, road detection, automatic emergency braking, night vision systems. Airbags, seat belt tightening system, collapsible steering column and pedestrian and child protection systems, hazard analysis and risk assessment.

**UNIT-IV:**

**ADAS in Autonomous Vehicles:** Levels of autonomous vehicles, system architecture, role of ADAS in path and decision making, V2V and V2I communications for collision avoidance and traffic management, vehicle fleet management and intelligent speed adoption, FPGA technology – benefits and applications, understanding ISO 26262 and its relevance to autonomous vehicles.

**UNIT-V:**

**Testing and Validation:** Testing methods - Simulation, hardware-in-the-loop and software-in-the-loop testing, real world testing - challenges and methodologies, key challenges in testing. Legal issues and regulatory challenges, industry trends and innovations in ADAS.

**TEXT BOOKS:**

1. Autonomous Driving and Advanced Driver-Assistance Systems (ADAS), Amit Kumar Mondal, Lentin Joseph, 1st edition, CRC Press, 2021
2. Advanced Driver Assistance Systems, Steve Zack, Kurt Shadbolt, Scott Brown, 1st edition, Goodheart-Wilcox Publisher, 2023

**REFERENCES:**

1. Intelligent Vehicle Technologies: Theory and Applications, Vlacic, Michel Parent and Fumio Harashima, Butterworth-Heinemann publications, 2001
2. Autonomous Vehicle Technology-A Guide for Policymakers, James M. Anderson, Nidhi Kalra, Karlyn D. Stanley, Paul Sorensen, Constantine Samaras, Oluwatobi A. Oluwatola, RAND Corporation, 2016
3. Autonomous Driving - Technical, Legal and Social Aspects, J. Christian Gerdes, Barbara Lenz, Hermann Winner, Springer, 2016

**ONLINE RESOURCES:**

1. <https://www.udemy.com/course/advanced-driver-assistance-systems>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22PC1AE402) AUTOMOTIVE EMBEDDED SYSTEMS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Electrical and Electronics Engineering

#### COURSE OBJECTIVES:

- To outline the evolution and role of automotive embedded systems in sustainable mobility
- To understand the technologies and standardization of electronic control units in modern vehicles
- To learn the concepts of ECU design, testing and validation through hardware-in-the-loop simulation
- To study advanced ECU systems and their applications in automobiles
- To know the communication protocols and the implementation of X by Wire technologies

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explore the evolution and role of automotive embedded systems in sustainable mobility

**CO-2:** Explain the technologies and standardization of electronic control units in modern vehicles

**CO-3:** Discuss the concepts of ECU design, testing and validation through hardware-in-the-loop simulation

**CO-4:** Describe the advanced ECU systems and their applications in automobiles

**CO-5:** Apply the communication protocols and the implementation of X by Wire technologies

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | 1    | -    | 3    | 1    | 1    | 2    | 1    | 1     | -     | 2     | 3                               | 3     |
| CO-2 | 3                     | 1    | 1    | -    | 3    | 1    | 1    | 2    | 1    | 1     | -     | 2     | 3                               | 3     |
| CO-3 | 3                     | 1    | 1    | -    | 3    | 1    | 1    | 2    | 1    | 1     | -     | 2     | 3                               | 3     |
| CO-4 | 3                     | 1    | 1    | -    | 3    | 1    | 1    | 2    | 1    | 1     | -     | 2     | 3                               | 3     |
| CO-5 | 3                     | 1    | 1    | -    | 3    | 1    | 1    | 2    | 1    | 1     | -     | 2     | 3                               | 3     |

#### UNIT-I

**Introduction to Automotive Embedded Systems:** Historical overview of electronics in automobiles, classification of Automotive embedded systems, evolution of automotive embedded systems, successes and failures in automotive electronics. Evolving smart vehicles with ECU. Role of Embedded system in sustainable mobility solutions.

## **UNIT-II**

**ECU technologies:** Components, configuration and diversification of electronics. Introduction to microcomputers, software complexity, algorithms, hierarchy design and standardization.

## **UNIT-III**

**ECU Design:** Design principles of ECUs, reliability assessment, and quality control. Introduction to Hardware-in-the-Loop (HIL) simulation for ECU testing and validation. Wire harness design, challenges and optimization strategies.

## **UNIT-IV**

**Advanced ECU Systems and Applications:** Engine ECU, Chassis ECU systems for vehicle dynamics, ABS, steering, and adaptive cruise control. Body and lighting ECU systems including airbags and keyless entry.

## **UNIT-V**

**CAN protocol:** Communication fundamentals – frequency, bandwidth, power measurement, signal to noise ratio, wired communication – network comparison, two tier approach, CAN applications and LIN applications. Challenges and opportunities of X-by-wire system and design requirements, steer-by-wire, brake-by-wire, electronic throttle including adaptive cruise control, shift-by-wire.

## **TEXT BOOKS:**

1. Automotive Embedded Systems: Key Technologies, Innovations, and Applications, M. Kathires, R. Neelaveni, 1st Edition, Springer, 2023
2. Understanding Automotive Electronics, William B. Ribbens, 6th Edition, Butterworth-Heinemann, 2003
3. Embedded Systems – A Contemporary Design Tool, James K. Peckol, John Wiley, 2008

## **REFERENCES:**

1. Automotive Embedded Systems Handbook, Nicolas Navet and Françoise Simonot-Lion, CRC Press, 2009
2. Encyclopedia of Automotive Engineering, David Crolla, David E. Foster, Toshio Kobayashi, Nicholas Vaughan, 1st Edition, Wiley, 2015
3. Embedded System Design: An Introduction to Processes, Tools, and Techniques Arnold Berger, CMP Books, 2001
4. An Embedded Software Primer, David E. Simon, Pearson Education, 2001
5. Computers as Components, Wayne Wolf, Morgan Kaufmann, 2001

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PE1AE401) ARTIFICIAL INTELLIGENCE FOR AUTONOMOUS AND CONNECTED VEHICLES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Elements of Automobile Engineering, Automotive Chassis, Automotive Engines, Automotive Electrical and Electronics

#### COURSE OBJECTIVES:

- To understand the trends and challenges of using artificial intelligence in autonomous vehicles
- To explore the state-of-the-art artificial intelligence architecture in autonomous vehicles
- To study the usage of artificial intelligence in autonomous car driving assistance systems
- To employ artificial intelligence in recognizing the driver's facial expressions
- To address security and privacy concerns of using artificial intelligence in automobiles

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the trends of artificial intelligence and their applications in autonomous vehicles

**CO-2:** Examine functional and technological aspects of autonomous vehicle architecture

**CO-3:** Apply artificial intelligence to the driver assistance system

**CO-4:** Implement artificial intelligence techniques to recognize driver emotions in autonomous vehicles

**CO-5:** Propose machine learning-based solutions to mitigate cyber-attacks and ensure the security of artificial intelligence driven autonomous vehicles

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 1                     | 2    | 3    | 2    | 2    | 2    | 2    | 1    | 1    | 1     | -     | 2     | 3                               | 2     |
| CO-2 | 3                     | 3    | 3    | 2    | 3    | 2    | 2    | 2    | 1    | 3     | -     | 2     | 3                               | 3     |
| CO-3 | 3                     | 3    | 3    | 3    | 3    | 2    | 2    | 2    | 2    | 3     | -     | 2     | 3                               | 3     |
| CO-4 | 3                     | 3    | 3    | 3    | 3    | 3    | 3    | 2    | 2    | 3     | -     | 2     | 3                               | 3     |
| CO-5 | 3                     | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 2    | 3     | -     | 2     | 3                               | 3     |

#### UNIT-I:

**Artificial Intelligence in Autonomous Vehicles – Trends and Challenges:** Research trends of artificial intelligence for autonomous vehicles, vehicle detection, rear end collision avoidance, traffic signal and sign recognition, lane detection and tracking,

pedestrian detection, dataset of autonomous vehicles, current industry standards, challenges and opportunities.

**Revolution of Computational AI in Autonomous Vehicle Technology:** Phases of autonomy, learning methodologies, advancement in intelligent vehicles, classification of technological advances in vehicle technology, vehicle architecture adaptation and future directions of autonomous driving.

#### **UNIT-II:**

**State of the Art AI Approaches toward Driverless Technology:** Role, history of AI in driverless cars, advancements in technology, algorithms and case study on Tesla.

Architectural Evolution of Autonomous Vehicles: Technology used in autonomous vehicles, analysis of autonomous vehicle architecture, functional architecture and challenges in building the architecture of autonomous vehicles.

#### **UNIT-III:**

**Autonomous Cars Driver Assistance System:** Traffic video surveillance, intelligent driver assistance system, traffic police hand gesture identification, vehicle brake and indicator light identification and collision avoidance for autonomous cars in dynamic and static environments.

#### **UNIT-IV:**

**Recognizing Driver's Emotions:** Models for driver assistance systems, face detection, facial emotion recognition – methods, dataset, preprocessing and grayscale equalization.

#### **UNIT-V:**

**Security and Privacy Issues:** Threat models for autonomous vehicles with AI in IoT enabled environments, physical attacks using AI, cybersecurity issues for autonomous vehicles, cyberattack defence mechanisms and solutions based on machine learning.

#### **TEXTBOOK:**

1. Artificial Intelligence for Autonomous Vehicles, Sathiyaraj Rajendran, 1st Edition, WILEY Publishers, 2024

#### **REFERENCE:**

1. AI – enabled Technologies for Autonomous and Connected Vehicles, Yi Lu Murphey, 1st Edition, SPRINGER, 2023

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

### (22PE1AE402) TWO AND THREE-WHEELER TECHNOLOGY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis and Automotive Engines

#### COURSE OBJECTIVES:

- To understand the fundamental concepts of two and three-wheelers
- To study the design trends of the power train
- To analyze the functionality of transmission systems
- To explore the steering, suspension and brake systems in two and three-wheelers
- To outline the road performance and maintenance of two and three-wheelers

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Illustrate the concepts and layouts available in two and three-wheelers

**CO-2:** Explain the power train of two and three-wheelers

**CO-3:** Describe the working of two and three-wheeler transmission system

**CO-4:** Discuss about the steering, suspension and brake system of two and three-wheelers

**CO-5:** Examine the road performance and maintenance of two and three-wheelers

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 2    | 1    | 2    | 2    | 1    | 1    | 1    | 2     | 2     | 2     | 3                               | 2     |
| CO-2 | 3                     | 3    | 3    | 2    | 3    | 2    | 2    | 1    | 1    | 2     | 2     | 2     | 3                               | 3     |
| CO-3 | 3                     | 3    | 3    | 2    | 3    | 2    | 2    | 1    | 1    | 2     | 2     | 2     | 3                               | 3     |
| CO-4 | 3                     | 3    | 3    | 2    | 3    | 3    | 2    | 1    | 2    | 3     | 2     | 3     | 3                               | 3     |
| CO-5 | 3                     | 3    | 3    | 3    | 3    | 3    | 3    | 1    | 2    | 3     | 3     | 3     | 3                               | 3     |

#### UNIT-I:

**Introduction:** Development, classification, layouts of two wheelers, motorcycles, scooters, mopeds and three wheelers, applications & capacity – goods and passengers and study of technical specification of two and three-wheelers.

**Frames & Body:** Types of frames, construction, loads, design consideration, materials, types of three-wheeler bodies, layout, aerodynamic, aesthetics and ergonomic considerations for bodywork.

#### UNIT-II:

**Powertrain:** Selection of engines for two and three-wheelers, design considerations for two and three-wheeler power plants, recent engine developments and electric two and three wheelers.

**UNIT-III:**

**Transmission System:** Clutch – Different types of clutches used in two and three wheelers, types of transmissions - gear shift mechanism, belt transmission, continuous variable transmission, cush drive, final drive, differential for three-wheeler and wheel drive arrangement.

**UNIT-IV:**

**Steering, Suspension and Brake System:** Steering – Steering geometry, steering column construction and steering system for three wheelers. Suspension – Suspension requirements, design considerations, trailing and leading link, swinging arm, springs and shock absorbers. Brakes - hand and foot operated. Wheel types - spokes, disc, split and special tire requirements for two & three wheelers.

**UNIT-V:**

**Road Performance:** Handling characteristics, driver and seating arrangement, ergonomics and comfort, road holding and vehicle stability, riding characteristics, safety arrangements and racing bikes – requirements.

**Maintenance:** Preventive and brake down maintenance, factors affecting fuel economy and emission.

**TEXT BOOKS:**

1. Motorcycle Design and Technology, Gaetano Cocco, Giorgio Nada Editore, 2013
2. Two and Three-Wheeler Technology, Dhruv U. Panchal, PHI Learning Pvt. Ltd. 2015

**REFERENCES:**

1. Motorcycle: Evolution, Design and Passion, Mick Walker, Johns Hopkins, 2006
2. Encyclopedia of Motor Cycling, 20 Volumes, Marshall Cavensih, New York and London, 1989
3. Motorcycle Tuning: Chassis, John Robinson, Butterworth-Heinemann, 2001
4. Service Manuals of Manufacturers of Indian Two & Three wheelers

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1AE403) ADVANCED MANUFACTURING TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Manufacturing Technology, Engineering Metallurgy and Materials, Manufacturing of Automotive Components

#### COURSE OBJECTIVES:

- To explore various non-traditional machining processes, parameters and applications
- To understand the principles of thermal energy-based material removal processes
- To outline the principles of spark energy-based material removal processes
- To know the principles of chemical and electro-chemical-based material removal processes
- To introduce Industry 4.0 and various additive manufacturing processes

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Classify the machining methods and applications

**CO-2:** Interpret the process parameters to be adopted and applicability of various materials that are suitable for thermal energy

**CO-3:** Review the process parameters to be adopted and applicability of spark energy-based machining processes

**CO-4:** Identify the process parameters to be adopted and applicability of various materials that are suitable for chemical and electro-chemical energy-based machining processes

**CO-5:** Explain Industry 4.0 and various additive manufacturing processes

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 1    | 2    | -    | 2    | 2    | 1    | 1    | 2     | -     | 1     | 3                               | 2     |
| CO-2 | 3                     | 2    | 1    | 2    | -    | 2    | 2    | 1    | 1    | 2     | -     | 1     | 3                               | 2     |
| CO-3 | 3                     | 2    | 1    | 2    | -    | 2    | 2    | 1    | 1    | 2     | -     | 1     | 3                               | 2     |
| CO-4 | 3                     | 2    | 1    | 2    | -    | 2    | 2    | 1    | 1    | 2     | -     | 1     | 3                               | 2     |
| CO-5 | 3                     | 2    | 1    | 2    | 2    | 2    | 2    | 1    | 1    | 2     | -     | 1     | 3                               | 2     |

#### UNIT-I:

**Introduction:** Unconventional machining processes, need and classification.

**Mechanical Energy Based Processes:** Abrasive Jet Machining (AJM), Water Jet Machining (WJM), Abrasive Water Jet Machining (AWJM), Ultrasonic Machining (USM). Working principles, equipment used, Process parameters, Material removal Rate (MRR) and applications.

**UNIT-II:**

**Thermal Energy Based Processes:** Laser Beam Machining (LBM), Plasma Arc Machining (PAM) and Electron Beam Machining (EBM) – principles, equipment, process parameters and applications.

**UNIT-III:**

**Electrical Energy Based Processes:** Electric Discharge Machining (EDM) - working principle, equipment, process parameters, MRR, electrode materials, power circuits, dielectric fluids, flushing, applications and Wire Cut EDM – applications.

**UNIT-IV:**

**Chemical and Electro-Chemical Energy Based Processes:** Chemical Machining (CHM) – etchants, maskants, techniques of applying maskants, process parameters, MRR, applications and Electro Chemical Machining (ECM) – principles, equipment, MRR, processes parameters and applications.

**UNIT-V:**

**Industry 4.0:** Introduction to cyber physical systems, next generation sensors and industrial internet of things.

**Additive Manufacturing:** Introduction, classification, comparison between Additive manufacturing and conventional manufacturing and advantages. Overview on binder jetting, directed energy deposition, material extrusion, material jetting, powder bed fusion, sheet lamination and vat photopolymerization processes.

**TEXT BOOKS:**

1. Advanced Machining Processes, Vijay K. Jain, 2<sup>nd</sup> Revised Edition, Allied Publishers, 2021
2. Modern Machining Processes, P. C. Pandey, H. S. Shan, Tata McGraw-Hill Education, 2017
3. Additive Manufacturing Technologies: Rapid Prototyping to Direct Digital Manufacturing, Ian Gibson, David W. Rosen, Brent Stucker, Springer, 2010

**REFERENCES:**

1. Nontraditional Manufacturing Processes, Benedict G. F, Marcel Dekker, CRC Press, 2019
2. Advanced Methods of Machining, McGeough, Chapman and Hall, London, 1998
3. Unconventional Machining Processes, P. K. Mishra, Narosa, Publishing House, 1997
4. Rapid Prototyping: Principles & Applications, Chuah Chee Kai, Leong Kah Fai, World Scientific, 2010
5. The Concept Industry 4.0: An Empirical Analysis of Technologies and Applications in Production Logistics, Christoph Jan Bartodziej, Springer Gabler Wiesbaden, 2017

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1ME401) COMPUTATIONAL FLUID DYNAMICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fluid Mechanics, Heat Transfer

#### COURSE OBJECTIVES:

- To familiarize with the differential equations that form basis for CFD
- To understand different numerical methods involved in problem solving
- To formulate different kinds of physical problems with the different schemes and boundary conditions
- To apply the CFD techniques to physical problems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop the fluid flow and heat transfer governing equations

**CO-2:** Apply and analyze different mathematical models and computational methods for flow and heat transfer simulations

**CO-3:** Conduct stability analysis and check the applicability of different schemes

**CO-4:** Solve computational problems related to fluid flow and heat transfer with FDM and FVM

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | -    | 3    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 3     |
| CO-2 | 3                     | 3    | 2    | -    | 3    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 3     |
| CO-3 | 3                     | 3    | 3    | -    | 2    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 3     |
| CO-4 | 3                     | 3    | 3    | -    | 2    | -    | -    | -    | -    | 2     | 1     | 2     | 3                               | 3     |

#### UNIT-I:

**Introduction to Computational Fluid Dynamics:** Computational Fluid Dynamics: Computational Fluid Dynamics as a Research Tool, Computational Fluid Dynamics as a Design Tool, The Impact of Computational Fluid Dynamics - some other examples: Automobile and Engine Applications, Industrial Manufacturing Applications, Civil Engineering Applications, Environmental Engineering Applications, Naval Architecture Application.

#### UNIT-II:

**The Governing Equations of Fluid Dynamics:** Introduction, Models of the Flow – Finite Control Volume, Infinitesimal Fluid Element, The Substantial Derivative (Time Rate of Change Following a Moving Fluid Element), The Divergence of the Velocity – Its Physical Meaning. The Continuity Equation – All Four models of flow, Integral versus Differential form of the Equations, The Momentum Equation, The Energy Equation, Equations for viscous flow (the Navier – stokes equation), Equations for Inviscid Flow

(the Euler Equations), Physical Boundary Conditions. Mathematical Behavior of Partial Differential Equations, Classification of PDEs.

#### **UNIT-III:**

**Finite Difference Method:** Taylor's series – Derivation of Finite Difference Formulae for Partial Derivative Terms - FD formulation of 1D Elliptic PDEs - 1D steady state heat transfer problems – Cartesian co-ordinate systems – boundary conditions. Explicit and Implicit Approaches – errors and analysis of stability

#### **UNIT-IV:**

**Finite Volume Method:** Finite Volume Method – Formation of Basic rules for Finite Volume approach – General Nodal Equation – Interface Thermal Conductivity – Treatment of Source Term and Treatment of Nonlinearity. Solution of 1D Elliptic PDEs - Heat conduction problems – Solution of 1D Parabolic PDEs – Explicit Method and Implicit Methods

#### **UNIT-V:**

**FVM to Convection and Diffusion:** General Form of Governing Equations for Fluid Flow and Heat transfer – Burger's equation - Steady 1D Convection Diffusion – Discretization Schemes and their assessment – Treatment of Boundary Conditions - Pressure Velocity Coupling - SIMPLE & SIMPLER (revised algorithm) Algorithms.

#### **TEXT BOOKS:**

1. Computational Fluid Dynamics: Basics with Applications, John D. Anderson, McGraw-Hill, 2015
2. Numerical Heat Transfer and Fluid Flow, Suhas V. Patankar, Hemashava Publishers Corporation & McGraw-Hill, 2018

#### **REFERENCES:**

1. Finite Difference Methods in Heat Transfer, Özişik, M. Necati, Helcio R. B. Orlande, Marcelo J. Colaço, and Renato M. Cotta, CRC press, 2017
2. Heat Transfer: A Basic Approach, Ozisik, M. Necati, McGraw-Hill, 1985
3. Fundamentals of Computational Fluid Dynamics, Tapan K. Sengupta, Universities Press, 2004
4. An Introduction to Computational Fluid Dynamics, H. K. Versteeg and W. Malalasekera, Pearson Education, 2007
5. Computational Fluid Flow and Heat Transfer, Muralidaran, Narosa Publication, 2014

#### **ONLINE RESOURCES:**

1. <https://nptel.ac.in/courses/112105045>
2. <https://ocw.mit.edu/courses/2-29-numerical-fluid-mechanics-spring-2015/>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PE1ME412) DESIGN FOR MANUFACTURING AND ASSEMBLY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Manufacturing Technology, Manufacturing of Automotive Components, Design of Automotive Components

#### COURSE OBJECTIVES:

- To impart the knowledge on steps involved in design process and material selection
- To understand about the design rules involved in machining and casting
- To understand about the design rules involved in metal joining, forging, extrusion and sheet metal work
- To understand about the design principles involved in manual and automatic assembly transfer systems

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Apply the knowledge on steps involved in design process and material selection

**CO-2:** Apply the knowledge on design rules involved in machining and casting

**CO-3:** Analyze the design rules involved in metal joining, forging, extrusion

**CO-4:** Design and analyze the principles involved in manual and automatic assembly transfer systems

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 2                     | 2    | 2    | 1    | -    | 3    | 3    | 1    | 3    | -     | -     | -     | 3                               | 1     |
| CO-2 | 2                     | 2    | 2    | 1    | -    | -    | -    | -    | 3    | -     | -     | -     | 3                               | 1     |
| CO-3 | 2                     | 2    | 2    | 1    | -    | -    | -    | -    | 3    | -     | -     | -     | 2                               | 1     |
| CO-4 | 2                     | 2    | 2    | 1    | -    | 3    | 3    | 1    | 3    | -     | -     | -     | 3                               | 1     |

#### UNIT-I:

**Introduction:** Design philosophy, steps in design process, general design rules for manufacturability and basic principles of designing for economical production.

**Creativity in Design:** Design aspects covering environmental concerns and sustainability, power consumption, operational safety and fool proofing.

**Materials:** Selection of materials for design, commonly used metal sections, criteria for material selection, material selection interrelationship with process selection and process selection charts.

#### UNIT-II:

**Machining Process:** Overview of various machining processes, general design rules for machining, dimensional tolerance and surface roughness, design for machining ease,

redesigning of components for machining ease with suitable examples and general design recommendations for machined parts.

**Metal Casting:** Appraisal of various casting processes, selection of casting process, general design considerations for casting, casting tolerances, use of solidification simulation in casting design and product design rules for sand casting.

#### **UNIT-III:**

**Metal Joining:** Appraisal of various welding processes, factors in the design of weldments, general design guidelines - pre and post treatment of welds, effects of thermal stresses in weld joints and design of brazed joints.

**Forging and Extrusion:** Design factors for forging, closed die forging design, parting lines of die drop forging die design, general design recommendations and design guidelines for extruded sections.

#### **UNIT-IV:**

**Assembly Advantages:** Development of the assembly process, choice of assembly method, assembly advantages, social effects of automation and overview of design for additive manufacturing.

**Automatic Assembly Transfer Systems:** Continuous transfer, intermittent transfer, indexing mechanisms and an operator - paced free transfer machine.

#### **UNIT-V:**

**Design of Manual Assembly:** General design guidelines for manual assembly, development of the systematic DFA methodology, assembly efficiency, classification system for manual handling, classification system for manual insertion and fastening, effect of part symmetry on handling time, effect of part thickness and size on handling time and effect of weight on handling time.

#### **TEXT BOOKS:**

1. Assembly Automation and Product Design, Geoffrey Boothroyd, 2<sup>nd</sup> Edition, CRC Press, 2005
2. Engineering Design – Material and Processing Approach, George E. Dieter, 3<sup>rd</sup> Edition, McGraw-Hill, 2000
3. Material Selection in Mechanical Design, Michael F. Ashby, 4<sup>th</sup> Edition, Butterworth-Heinemann, 2010

#### **REFERENCES:**

1. Handbook of Product Design, Geoffrey Boothroyd, 2<sup>nd</sup> Edition, Marcel and Dekker, 2002
2. Computer Aided Assembly Planning, A. Delchambre, Springer, 1992

#### **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/107/103/107103012/>
2. <https://archive.nptel.ac.in/courses/112/101/112101005/>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PC2AE402) AUTOMOTIVE EMBEDDED SYSTEMS LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Electrical and Electronics Engineering

#### COURSE OBJECTIVES:

- To equip skills to implement model-based development control strategies and closed-loop control techniques
- To provide hands-on experience with automotive sensors, actuators, and electronic control systems
- To operate xEV mode controls for modern hybrid and electric vehicles while troubleshooting CAN-based communication between ECUs
- To familiarize the use of electronic control units for flashing/programming, data logging, and data analysis
- To provide practical exposure to EV motor performance and battery charging/discharging characteristics

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Develop model-based control strategies for electronic throttle and motor systems, including PID and PWM calibration, based on vehicle parameters

**CO-2:** Apply xEV mode controls (ICE, EV, and hybrid modes) and closed-loop techniques

**CO-3:** Configure and troubleshoot ECU-to-ECU communication using CAN protocol and analyze logged data for efficient system operation

**CO-4:** Create ECU programming files, flash ECUs, and evaluate system performance through parameter measurement and data analysis

**CO-5:** Test the performance of EV motors and charging and discharging characteristics of batteries

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | 3    | 2    | 2    | -    | -    | 2    | 2    | -     | 3     | 3     | 2                               | 3     |
| CO-2 | 3                     | 3    | 3    | 2    | 3    | -    | -    | 2    | 2    | 2     | 3     | 3     | 3                               | 3     |
| CO-3 | 3                     | 2    | 2    | 3    | 3    | -    | -    | 2    | 2    | 3     | 3     | 3     | 3                               | 3     |
| CO-4 | 3                     | 3    | 3    | 3    | 2    | -    | -    | 1    | 1    | 3     | 3     | 1     | 3                               | 3     |
| CO-5 | 3                     | 3    | 3    | 3    | 2    | -    | -    | 1    | 1    | 3     | 3     | 1     | 3                               | 3     |

## **LIST OF EXPERIMENTS:**

### **ANY 10 EXPERIMENTS to be conducted from the following**

1. Electronic throttle control based on engine speed and engine torque
2. Electric motor control based on vehicle speed and accelerator pedal
3. xEV modes - IC engine mode control, EV mode control and hybrid mode control
4. Cooling fan operation based on coolant temperature
5. CAN communication basics and ECU to ECU communication
6. Glow plug operation based on engine speed and engine torque
7. Electric turbocharger ON/OFF control
8. Closed-loop electronic throttle control with PID calibration
9. Close-loop mass air flow control using electronic throttle body
10. Performance analysis of BLDC motor
11. Performance analysis of PMSM motor
12. Obtain characteristics of the charging and discharging of lithium –ion batteries

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PC2AE411) VEHICLE MAINTENANCE AND TESTING LABORATORY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 0               | 2   | 1 |

| EVALUATION SCHEME |    |    |    |     |       |
|-------------------|----|----|----|-----|-------|
| D-D               | PE | LR | CP | SEE | TOTAL |
| 10                | 10 | 10 | 10 | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis, Automotive Engines, Automotive Electrical and Electronics, Vehicle Dynamics, Electric and Hybrid Vehicles

#### COURSE OBJECTIVES:

- To provide hands-on training on vehicle inspection
- To demonstrate wheel balancing and alignment tests
- To conduct vehicle pollution test
- To diagnosis and troubleshoot the vehicle using suitable test equipment

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Examine the vehicle and engine subsystems

**CO-2:** Experiment wheel balancing and steering

**CO-3:** Analyze vehicle emission

**CO-4:** Diagnosis and troubleshoot the vehicle

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | -    | 3    | 1    | 2    | 3    | -    | 3    | 2     | 1     | 3     | 3                               | 2     |
| CO-2 | 3                     | 1    | -    | 3    | 1    | 2    | 3    | -    | 3    | 2     | 1     | 3     | 3                               | 2     |
| CO-3 | 3                     | 1    | -    | 3    | 3    | 2    | 3    | -    | 3    | 2     | 1     | 3     | 3                               | 2     |
| CO-4 | 3                     | 1    | -    | 3    | 3    | 2    | 2    | -    | 3    | 2     | 1     | 2     | 3                               | 1     |

#### LIST OF EXPERIMENTS:

##### ANY 10 EXPERIMENTS to be conducted from the following

1. Vehicle inspection
2. Engine compression test
3. Engine manifold vacuum test
4. Ignition timing test
5. Automotive battery test
6. Wheel balancing of wheel and tyre assembly
7. Headlight alignment test
8. Petrol vehicle exhaust analysis
9. Diesel smoke measurement
10. Study of wheel geometry and Ackerman steering mechanism
11. On-board diagnostic test.

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AE404) ELECTRIC VEHICLE CHARGING INFRASTRUCTURE TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Electricals and Electronics, Automotive Electrical and Electronics

#### COURSE OBJECTIVES:

- To understand the architecture and classification of EV charging infrastructure
- To analyze the components and design of fast charging stations
- To explore power electronics in EV chargers
- To study various charging strategies
- To evaluate business models and smart charging strategies

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain EV architecture and charging infrastructures, including on-board, off-board and wireless chargers

**CO-2:** Analyze fast charging station design, components, and renewable energy integration

**CO-3:** Understand power electronics devices and their role in EV charging systems.

**CO-4:** Describe various charging strategies

**CO-5:** Evaluate business models, tariff structures, and smart charging strategies like V2G and V2H

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | -    | 1    | -    | 3    | 2    | 2    | -    | -    | -     | -     | 2     | 3                               | 2     |
| CO-2 | 3                     | -    | 3    | -    | 1    | 2    | 2    | -    | -    | -     | -     | 3     | 3                               | 2     |
| CO-3 | 2                     | -    | 1    | -    | 1    | 2    | 2    | -    | -    | -     | -     | 3     | 3                               | 2     |
| CO-4 | 1                     | -    | 2    | -    | 1    | 2    | 2    | -    | -    | -     | -     | 3     | 3                               | 2     |
| CO-5 | 3                     | -    | 1    | -    | 3    | 2    | 3    | -    | -    | -     | -     | 3     | 3                               | 2     |

#### UNIT-I:

**Standards and Specifications:** Architecture of EV, classification of charging Infrastructure, on board chargers, off board chargers, wireless chargers, charging standards – SAE J1772 AC/DC charging, CHAdeMO, CCS, GB/T, Tesla, IEC 162191, Electric Vehicle Supply Equipment (EVSE), EVSE connector specifications and requirements for fast-charging station.

#### UNIT-II:

**Fast-Charging Station:** Single Line Diagram (SLD) of fast charging station, elements of fast charging station, power grid, power cables- IEC 62893, switchgear, transformer,

energy meters and power quality meters, fast chargers, plugs and connectors, IEC 61851 methods, DC connectors, CCS, CHAdeMO, GB/T, NACS, fast charging station using renewable power sources, digital communication for fast charging station and fast charging - case study in India.

#### **UNIT-III:**

**Power Electronic Preliminaries for EV Chargers:** Power devices, diodes – MOSFET, IGBT, AC-DC converter, diode bridge rectifier, non-isolated DC-DC converter, boost converter, and DC-AC converter.

**On Board Chargers (OBC):** Two stage OBC, bridge boost power factor correction circuit (boost PFC), DC-DC interface converter, phase shift isolated full bridge converter – principle of operation, case study of 3.3 kW two-stage OBC – Specifications and introduction to DC fast chargers.

#### **UNIT-IV:**

**Types of Battery Charging:** Charging methods – constant voltage current charging, constant current multilevel charging, incremental method, pulse charging algorithm, sinusoidal pulse charging, temperature and heat control

**Wireless Power Transfer:** Fundamentals of wireless power transfer (WPT) technology, classification of WPT-based battery chargers, inductive power transfer (IPT) based EV charging – block diagram, coupling coils, comparison, case study - 3.3 kW WPT specification and block diagram.

#### **UNIT-V:**

**Business Model and Tariff Structure for Fast-Charging Station:** Business models, integrated model, independent model, expenses, battery swapping and tariff structures

**Smart Charging:** V2G charging system – design, control strategy, case study, V2B, V2H, comparison and V2V charging.

#### **TEXT BOOKS:**

1. Fast-Charging Infrastructure for Electric and Hybrid Electric Vehicles, P. Sivaraman, Sharmeela, Sanjeevikumar P., IEEE Wiley Press, 2023
2. Power Electronics for Electric Vehicles and Energy Storage, Dharavath Kishan, Ramani K., B. Dastagiri R., and P. Prabhakaran, CRC Press, 2023

#### **REFERENCES:**

1. Smart Charging Solutions for Hybrid and Electric Vehicles, Sulabh Sachan, Sanjeev Kumar, Wiley Press, 2022
2. Fast Charging and Resilient Transportation Infrastructures in Smart Cities, Hossam A. Gabbar, Springer, 2022
3. Power Electronics Devices, Circuits, and Applications, Muhammad H. Rashid, Pearson, 2014

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PE1AE405) AUTOMOTIVE NOISE AND VIBRATION CONTROL

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Vehicle Dynamics, Automotive Chassis

#### COURSE OBJECTIVES:

- To understand causes and impact of the automotive noise vibration and harshness
- To study the measurement techniques of vibration and noise
- To acquire knowledge on sources of noise, vibration and harshness
- To know the passive noise treatment
- To learn the noise and vibrations in automotives

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Recognize the impact of automotive noise, vibration and harshness

**CO-2:** Explain the various tools for measurement of vehicle vibration and noise

**CO-3:** Identify the sources of noise in automotives

**CO-4:** Apply the passive noise treatment techniques to control noise

**CO-5:** Discuss the various method to control interior noise of vehicle

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-2 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-3 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-4 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-5 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |

#### UNIT-I:

**Introduction To NVH:** Vibration of beams, plates and shells. Basics of sound propagation, quantification of sound, noise sources, pass-by noise limits, automotive NVH sources, interior noise of vehicles, sound quality, ride comfort, noise and vibration control in vehicles.

#### UNIT-II:

**Transducers and Measurement Techniques:** Transducers and exciters. Sound pressure, intensity and power measurement, sound level meters, noise dosimeters, analyzers, signal generators, equipment for data acquisition and digital signal processing. Calibration of measurement microphones, calibration of shock and vibration transducers, metrology and traceability of vibration and shock measurements.

**UNIT-III:**

**Noise Source Identification Techniques:** Frequency and order domain analysis, sound intensity and sound power mapping. Introduction to array techniques, acoustic holography and beam forming, standard methods for evaluating sound absorption coefficient and transmission loss, types of sound absorbers, prediction of transmission loss and flanking transmission, damping materials and their applications.

**UNIT-IV:**

**Passive Noise Treatments:** Ducts and Mufflers - types of mufflers, performance parameters - acoustics and backpressure. Reactive and absorptive silencers, overall design considerations. Acoustic material characterization, sound transmission, absorption and damping, behaviour of acoustic material with respect to sound absorption and transmission

**UNIT-V:**

**Interior Noise Of Automobiles And Modal Analysis:** Interior noise sources, structure borne noise, airborne noise, refinement techniques and sound insulation. Definition of modal properties, modal analysis theory, FEM and experimental modal analysis, excitation sources and applications of modal analysis.

**TEXT BOOKS:**

1. Vehicle Noise and Vibration Refinement, Xu Wang, Woodhead Publishing, 2010
2. Vehicle Refinement: Controlling Noise and Vibration in Road Vehicles, M. Harrison, Butterworth-Heinemann, 2004

**REFERENCES:**

1. Mechanical Vibrations and Noise Engineering, G. Ambekar, PHI Learning, 2006
2. Understanding Active Noise Cancellation, C. H. Hansen, CRC Press, 2003
3. Vibration Monitoring, Testing, and Instrumentation, C. W. de Silva, CRC Press, 2007
4. Engineering Noise Control: Theory and Practice, D. A. Bies and C. H. Hansen, CRC Press, 2009

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PE1AE406) FARM POWER AND MACHINERY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis, Automotive Engines

#### COURSE OBJECTIVES:

- To study various farm mechanization, tractors and their applications
- To understand the tillage, sowing and planting equipment
- To learn about the inter-cultivation equipment
- To identify the plant protection equipment
- To provide an overview of different power outlets of tractor

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain tractors and need for farm mechanization

**CO-2:** Discuss the types of tillage, sowing and planting equipment

**CO-3:** Describe the specific functions of inter-cultivation equipment

**CO-4:** Present the requirements and features of various plant protection equipment

**CO-5:** Illustrate power outlets of tractors and power tillers

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | -                     | 3    | 2    | 2    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-2 | -                     | 3    | 2    | 2    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-3 | -                     | 3    | 2    | 2    | -    | 2    | 1    | -    | 1    | 2     | -     | 2     | 3                               | 3     |
| CO-4 | -                     | 3    | 2    | 2    | -    | 2    | 1    | -    | 1    | 2     | -     | 2     | 3                               | 3     |
| CO-5 | -                     | 3    | 2    | 2    | -    | 2    | 1    | -    | 2    | 2     | -     | 2     | 3                               | 3     |

#### UNIT-I:

**Introduction to Farm Mechanization:** Status of farm mechanization in India, objectives of farm mechanization, benefits of farm mechanization, bottlenecks of Indian farm mechanization system and classification of farm machines.

**Tractors:** General description, specification, functions, light, medium and heavy-duty tractors.

#### UNIT-II:

**Tillage Equipment:** Tillage- primary and secondary tillage equipment, working principles, forces acting on tillage tools, field operation patterns, draft measurement of tillage equipment, earth moving equipment – construction, working principles of bulldozer, trencher and excavators.

**Sowing and Planting Equipment:** Seeding methods, seed drill - components, calibration, seed metering mechanisms, test for seed uniformity, pneumatic seed drill,

planters - potato planter, sugarcane planter, zero till drill, strip till drill, rice transplanters, vegetable transplanters and drum seeder.

#### **UNIT-III:**

**Inter-Cultivation and Miscellaneous Equipment:** Inter-cultivation equipment- introduction, types of weeders, hoes, puddler, bund former, ridger, post hole digger, brush cutter and laser leveller.

#### **UNIT-IV:**

**Plant Protection Equipment:** Basic terminologies related to plant protection, sprayers, dusters, types and working principles, spray characteristics, care, maintenance, fertilizer application equipment, manure application equipment, flame weeding, variable rate application of fertilizers and chemicals.

#### **UNIT-V:**

**Power Outlets of Tractor and Power Tiller:** Belt pulley, drawbar, power tiller construction and working, power transmission system.

#### **TEXT BOOK:**

1. Principles of Farm Machinery, R. A. Kepner, Bainer Roy, and E. C. Barges, 2<sup>nd</sup> Edition, CBS Publishers and Distributors, 2005

#### **REFERENCES:**

1. Principles of Agricultural Engineering (Vol. I), A. M. Michael and T. P. Ojha, 3<sup>rd</sup> edition, Jain brothers, 2011
2. Elements of Agricultural Engineering, Jagdishwar Sahay, 4<sup>th</sup> Edition, Standard Publishers and Distributors, 2015

#### **ONLINE RESOURCES:**

1. <http://ecoursesonline.iasri.res.in/course/view.php?id=12>

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1AE407) GREEN HYDROGEN TECHNOLOGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Engines

#### COURSE OBJECTIVES:

- To explore the significance and applications of green hydrogen
- To know various green hydrogen production technologies
- To outline various hydrogen storage and transport technologies
- To understand the different applications of green hydrogen
- To study the economic and environmental aspects of green hydrogen

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Describe various details about hydrogen

**CO-2:** Explain different types of green hydrogen production technologies

**CO-3:** Present different types of hydrogen storage and transport technologies

**CO-4:** Identify the different applications of green hydrogen

**CO-5:** Explore the various economic and environmental aspects of green hydrogen

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-2 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-3 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-4 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-5 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |

#### UNIT-I:

**Introduction:** Global energy scenario, need for sustainable energy solutions. Hydrogen – properties, types – brown, grey, blue and green. Applications - oil refining, fertilizer, ammonia and methanol production. Advantages and limitations, material compatibility, standards. Production – coal, biomass, natural gas. Usage in IC engines – combustion and emissions

#### UNIT-II:

**Green Hydrogen Production:** Electrolysis – alkaline, proton exchange membrane, and solid oxide electrolysis. Photoelectrochemical water splitting, thermochemical water splitting. Renewable energy integration – solar, wind and hydroelectric power for electrolysis.

**UNIT-III:**

**Storage and Transportation:** Compressed gas storage, material-based storage – metal hydrides, chemical carriers, stationary storage, pipelines and transportation methods – tankers and shipping, infrastructure requirements for hydrogen supply chains, distribution networks and challenges.

**UNIT-IV:**

**Green Hydrogen Applications:** Power generation – hydrogen fuel cells, Integration into the power grid. Transportation – hydrogen fuel cell vehicles, Infrastructure for hydrogen mobility. Industrial applications – Refining, ammonia production, steel making and emerging applications in various industries.

**UNIT-V:**

**Economic and Environmental Aspects:** Cost analysis of green hydrogen production, Market trends and future projections, Environmental impact – life cycle assessment, comparison with fossil fuels and other hydrogen types, government incentives and policies, international agreements and collaborations.

**TEXT BOOKS:**

1. Hydrogen Economy and Green Hydrogen Technologies, Yakson Bobby, Independently Published, 2024
2. Alternative Fuels, S. S. Thipse James, 1<sup>st</sup> Edition, Jaico Publishing House, 2020
3. Hydrogen Fuel: Production, Transport and Storage, Ram B. Guptha, CRC press, 2008

**REFERENCES:**

1. Alternative Fuels for IC Engines, Anand Krishnaswamy, 1<sup>st</sup> Edition, McGraw-Hill, 2024
2. The Hydrogen Revolution: A Blueprint for The Future of Clean Energy, Marco Alverá, Hodder Studio, 2021
3. Hydrogen Energy: Challenges and Solutions for a Cleaner Future, Bahman Zohuri, Springer, 2008

**ONLINE RESOURCES:**

1. [https://onlinecourses.nptel.ac.in/noc22\\_ch66/preview](https://onlinecourses.nptel.ac.in/noc22_ch66/preview)

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VII Semester

#### (22PE1AE408) AUTOMOTIVE PRODUCT DEVELOPMENT STRATEGIES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

#### COURSE OBJECTIVES:

- To understand the principles of product design
- To study the product planning process
- To explore the challenges and opportunities involved in developing global products
- To know the product life cycle management of global products
- To identify the organizational issues in product design and development

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the principles of new product development and product styling to manage risks and ensure quality

**CO-2:** Plan product designs by evaluating competing products and styling specifications.

**CO-3:** Identify challenges in global product development and solve issues using root cause analysis and advanced tools.

**CO-4:** Use PLM tools to manage product development processes

**CO-5:** Identify organizational issues in product development process

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 2                     | 3    | 3    | 2    | -    | 2    | 2    | -    | 2    | 3     | 2     | 3     | 3                               | 1     |
| CO-2 | 2                     | 3    | 3    | 3    | -    | 3    | 2    | -    | 2    | 3     | 3     | 3     | 3                               | 1     |
| CO-3 | 2                     | 2    | 3    | 3    | -    | 3    | 2    | -    | 2    | 3     | 3     | 3     | 3                               | 1     |
| CO-4 | 2                     | 3    | 3    | 3    | -    | 3    | 2    | -    | 2    | 3     | 3     | 3     | 3                               | 1     |
| CO-5 | 1                     | 1    | 1    | 2    | -    | 2    | 3    | -    | 3    | 3     | 3     | 3     | 3                               | 1     |

#### UNIT-I:

**Introduction to Product Design:** Introduction, principles of new product development, success and failure in new products, risk management, funnel and its stages, quality control of product development and meeting targets. The principles of product styling-virtual perception of product style, attractiveness and product styling process.

#### UNIT-II:

**Product Planning:** Product planning process, opportunities in product planning, competing product analysis, style planning, factors of contextual styling and Intrinsic styling and styling specifications.

**UNIT-III:**

**Global Products:** Importance, challenges and opportunities of global products, changes and complexity in global products, global product problems – addressing potential problems, multiple causes and its effects, root cause and network of causes and measures, everyday product problems and action. Changing roles of product organizations, increased regulation of product, better managed product, multiple new products and breakthrough computer aided product development.

**UNIT-IV:**

**PLM Enabling Global Products:** Product lifecycle management (PLM), key characteristics and functions, importance, benefits of PLM, metrics and targets of PLM, PLM applications – data/document management, part/product management, process/workflow management and program/project management.

**UNIT-V:**

**Organizational Issues in Product Design and Development:** Interactions across organizational functions within respondents' organization and industry, basic elements of the product development process, effectiveness of communication methods and organizational barriers.

**TEXT BOOKS:**

1. Global Products, John Stark, Springer-Verlag, 2010
2. Product Design, Mike Baxter, Stanley Thornes Publishers, 1999

**REFERENCES:**

1. Product Design and Development, Karl T. Ulrich and Steven D. Eppinger, Tata McGraw-Hill, 2003
2. Product Innovation, David L. Rainey, Cambridge University Press, 2005
3. Product Development, Anil Mital, Anoop Desai, Anand Subramanian and Aashi Mital, Butterworth-Heinemann, 2008
4. New Product Development, Michael Z. Brooke and William Ronald Mills, Jaico Publishing House, 2008

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VII Semester

### (22PE1AE409) ALTERNATIVE FUELS AND POLLUTION CONTROL

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Engineering Chemistry and Automotive Engines

#### COURSE OBJECTIVES:

- To explore various gaseous fuels for sustainable mobility
- To examine the benefits, production methods, and engine modifications required for using biofuels and their hazards
- To analyze the production processes, advantages, and emissions of synthetic alternative fuels
- To identify the sources of automotive emissions, their impact on the environment and human health, and applicable emission standards
- To present different emission control technologies for SI and CI engines and their effectiveness in reducing pollution

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Identify various gaseous fuels, production methods, storage, and applications

**CO-2:** Explain the benefits, engine modifications, and health hazards of biofuels

**CO-3:** Compare different synthetic alternative fuels, production and emission characteristics

**CO-4:** Assess vehicle emissions, their impact on health and the environment, and interpret emission standards

**CO-5:** Recommend appropriate emission control technologies for SI and CI engines to comply with environmental standards

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 2    | 3    | 1    | -    | 3    | 2    | 1    | -    | -     | -     | 3     | 3                               | 3     |
| CO-2 | 3                     | 2    | 3    | 3    | -    | 3    | 3    | 1    | -    | -     | -     | 3     | 3                               | 3     |
| CO-3 | 3                     | 3    | 1    | 2    | -    | 3    | 2    | 1    | -    | -     | -     | 3     | 3                               | 3     |
| CO-4 | 3                     | 2    | 3    | 2    | -    | 3    | 3    | 1    | -    | -     | -     | 1     | 3                               | 3     |
| CO-5 | 3                     | 3    | 3    | 3    | -    | 3    | 2    | 1    | -    | -     | -     | 3     | 3                               | 3     |

#### UNIT-I:

**Gaseous Alternative Fuels:** Hydrogen – production methods – electrolysis of water, catalytic steam reforming of natural gas, photochemical conversion, on-board storage – liquid hydrogen, metal hydride, advantages and disadvantages. Compressed Natural Gas – production, storage, advantages, disadvantages, dispensing, material compatibility for CNG and engine modifications. Liquefied

Natural Gas – production, advantages, disadvantages, material compatibility for LNG, storage, dispensers and LNG-to-CNG conversion system.

#### **UNIT-II:**

**Biofuels:** Biogas and Biomethane – composition, production, storage, dispensing system, advantages, hazards of biogas. Methanol – production, applications, advantages, disadvantages, health hazards and storage. Ethanol – production, storage, advantages, disadvantages, health hazards. Biodiesel – feedstock selection, production, storage, advantages, disadvantages, hazards and engine modifications for biodiesel.

#### **UNIT-III:**

**Synthetic Alternative Fuels:** Dimethyl Ether – production, applications, advantages, hazards, storage. Gas to Liquid – production, advantages. Coal to Liquid – production, advantages. Tyre pyrolysis oil – production and emissions.

#### **UNIT-IV:**

**Vehicle Emissions:** Types of pollutants – carbon monoxide, oxides of nitrogen, unburned hydrocarbon, soot and particulate, effects of principal pollutants on human health.

**Emission Standards:** Emission norms, BS VI emission standards for 2-wheeler vehicle, 3-wheeler vehicle, light duty vehicle and emission control - in use vehicles.

#### **UNIT-V:**

**Emission Control in SI Engines:** Engine design parameters, positive crankcase ventilation, exhaust gas recirculation, catalytic converters, gasoline particulate filters and on-board diagnostics.

**Emission Control in CI Engines:** Exhaust gas recirculation, turbocharging, selective catalytic reduction catalyst and diesel particulate filters.

#### **TEXT BOOKS:**

1. Alternative Fuels, S. S. Thipse, 6<sup>th</sup> Edition, Jaico Publishing House, 2015
2. Engine Emissions-Fundamentals and Advances in Control, Pundir B. P., 2<sup>nd</sup> Edition, Narosa Publishing House, 2017
3. Indian Emission Regulation Booklet, The Automotive Research Association of India (ARAI)-Pune, 2021

#### **REFERENCES:**

1. Alternative Fuels for IC Engines, Anand Krishnasamy, Saurabh K. Gupta, 1<sup>st</sup> Edition, McGraw-Hill, 2024
2. Handbook of Alternative Fuel Technologies, James G. Speight, Sudarshan K. Loyalka & Sunggyu Lee, 2<sup>nd</sup> Edition, Taylor & Francis CBS, 2018
3. Internal Combustion Engine Fundamentals, John Heywood, 1<sup>st</sup> Edition, McGraw-Hill Education, 2017

#### **ONLINE RESOURCES:**

1. <https://archive.nptel.ac.in/courses/112/103/112103262/>
2. <https://archive.nptel.ac.in/courses/103/105/103105110/>
3. [https://onlinecourses.nptel.ac.in/noc22\\_de02/preview](https://onlinecourses.nptel.ac.in/noc22_de02/preview)
4. <https://www.coursera.org/learn/energy-and-environment>

## VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

### B.Tech. VIII Semester

### (22PE1AE410) AUTOMOTIVE BIOMECHANICS

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis

#### COURSE OBJECTIVES:

- To study the concepts of biomechanics and modeling of human response
- To understand the biomechanics and injury criteria of vehicle occupants
- To know the limb biomechanics and pedestrian safety
- To use the restrain mechanisms for automotive safety
- To provide safety systems considering biomechanics for injury prevention

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the concepts of biomechanics and modelling of human response

**CO-2:** Describe the biomechanics of head and thorax

**CO-3:** Analyze limb biomechanics and pedestrian safety in vehicle crashes

**CO-4:** Discuss restraint and protective systems in automobiles

**CO-5:** Apply biomechanics for injury prevention of vehicle occupants

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | -                     | 2    | 1    | -    | -    | 3    | 3    | 1    | -    | -     | -     | 1     | 2                               | 1     |
| CO-2 | -                     | 2    | 1    | -    | -    | 3    | 3    | 1    | -    | -     | -     | 1     | 2                               | 1     |
| CO-3 | -                     | 2    | 1    | -    | -    | 3    | 3    | 1    | -    | -     | -     | 2     | 2                               | 1     |
| CO-4 | -                     | 2    | 1    | -    | -    | 3    | 3    | 1    | -    | -     | -     | 2     | 2                               | 1     |
| CO-5 | -                     | 2    | 1    | -    | -    | 3    | 3    | 1    | -    | -     | -     | 2     | 2                               | 1     |

#### UNIT-I:

**Biomechanics and Occupant Response:** Basics of biomechanics, relevance, human and dummy modelling, crash testing, NCAP ratings, injury classification, injury criteria, biomechanical limits on humans, internal injuries – concussion, spinal injuries and chest injuries.

#### UNIT-II:

**Biomechanics of Thorax/Abdomen:** Chest protection, basics of head injuries, restraint design, design for minimization of injury risk, rating, difference between male and female occupants and driver.

**Biomechanics of Head:** Head protection, head impact, head injury criteria, neck injury, high speed vertical loading and skull protection.

**UNIT-III:**

**Pelvic and Limb Biomechanics:** Pelvic injury criteria, leg and knee injury criteria and torso impact tests.

**Pedestrian Safety:** Pedestrian protection, pedestrian accident simulation tests, obstacle detection and protection zones.

**UNIT-IV:**

**Restraint and Protective System:** Comfort and ergonomics, crash testing – frontal collisions, lateral collisions, rear-end collisions, rollover, biomechanical injury data analysis and development of future vehicle safety features.

**UNIT-V:**

**Injury Prevention:** Prevention of head injuries, soft tissue neck injury, head restraint geometry, padding material, self-aligning head restraint, WHIP seat, damping seat side, WipGARD, prevention of pelvic injuries and lower extremities.

**TEXT BOOKS:**

1. Integrated Automotive Safety Handbook, Mark Gonter and Ulrich W. Seiffert, SAE International, 2013
2. Basic Biomechanics, Susan J. Hall, 9<sup>th</sup> Edition, McGraw-Hill Education, 2021

**REFERENCES:**

1. Medical Instrumentation–Application & Design, Webster J. G., 3<sup>rd</sup> Edition, John Wiley and Sons, 2003
2. Biomechanics-Principles and Applications, Schneck D. J. and Bronzino J. D., 2<sup>nd</sup> Edition, CRC Press, 2000
3. Fundamentals of Biomechanics, Duane Knudson, 2<sup>nd</sup> Edition, Springer, 2007

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AE411) SPECIAL PURPOSE VEHICLES

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis

#### COURSE OBJECTIVES:

- To study the functions and features of cranes and excavators
- To learn the functions and features of graders
- To write the functions and features of haulage vehicles and lift trucks
- To know the functions and features of rooters, scarifiers and scrapers
- To provide the functions and features of compaction and other special purpose vehicles

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the functionality of different types of cranes and excavators

**CO-2:** Discuss the types and functionality of graders

**CO-3:** Describe the functions and features of haulage vehicles and lift trucks

**CO-4:** Outline the usage and specifications of the rooters, scarifiers and scrapers

**CO-5:** Present the features of various special purpose vehicles

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 2                     | 3    | -    | -    | -    | 2    | 2    | -    | 1    | 2     | -     | 2     | 3                               | 2     |
| CO-2 | 2                     | 2    | -    | -    | -    | 2    | 2    | -    | 1    | 2     | -     | 2     | 3                               | 3     |
| CO-3 | 2                     | 3    | -    | -    | -    | 2    | 2    | -    | 1    | 2     | -     | 2     | 3                               | 3     |
| CO-4 | 2                     | 3    | -    | -    | -    | 3    | 2    | -    | 1    | 3     | -     | 3     | 3                               | 3     |
| CO-5 | 2                     | 3    | -    | -    | -    | 3    | 3    | -    | 2    | 3     | -     | 3     | 3                               | 3     |

#### UNIT-I:

**Cranes and Excavators:** General description, specifications, functions of cranes, mobile cranes with strut, cantilever type jibs, tractor towed and tractor mounted cranes. General description, specification and functions of excavators, classification based on attachments, face shovel, drag shovel, hoe, drag-line and grab or clam shell, advantages and limitations.

#### UNIT-II:

**Graders:** Description, specification of tractor towed graders and motor graders, classification and functions of graders, functional details of spreading, mixing, ditching, bank sloping, snow removal, stripping, scarifying, and finishing, elementary details of transmission system (coupling, clutches, gear box, driving axles, propeller

shafts), running gear and operating equipment air braking system; hydraulic system and its components, merits and limitations of graders.

#### **UNIT-III:**

**Haulage Vehicles and Lift Trucks:** General description, specification and functions, self-propelled and tractor towed haulage vehicles, dumpers – front tipping; trucks – rear tipping, tractor towed semi-trailers and trailers (rear and side tipping, bottom dumping). General description, specification and functions, forklift trucks, alternative front-end equipment (attachments) – Jib arm, shovel bucket, squeeze clamp, boom, fork extensions, barrel forks. Scissors lift trucks - applications in industry, advantages and disadvantages.

#### **UNIT-IV:**

**Rooters, Scarifiers and Scrapers:** General description, specification and functions, tractor towed rooters and scarifiers - heavy duty, light duty. General description, specification and functions of tractor towed and motorized scrapers, scraper work in cutting, cambering, side hill cutting, spreading on embankments, compaction of fill merits and demerits.

#### **UNIT-V:**

**Compaction Vehicles and other Special Purpose Vehicles:** General description, specification and functions, smooth wheeled rollers, pneumatic tired rollers, agricultural Rollers, sheep's foot rollers, vibrating compactors. General description, specification and functions, ambulance, oil tankers, surveillance vehicle, television recording mobile unit, reefer vehicle, double decker bus, vestibule bus and fire fighting vehicle.

#### **TEXT BOOKS:**

1. Construction Planning, Equipment and Methods, Peurifoy R. L., Tata McGraw-Hill, 2002
2. Off-Road Vehicles, Ian Graham, Heinemann Library, 2008

#### **REFERENCES:**

1. Terramechanics and Off-Road Vehicle Engineering, Wong J., Butterworth-Heinemann, 2009
2. Mechatronics and Intelligent Systems for Off-road Vehicles, Rovira Más, Francisco, Zhang, Qin, Hansen, Alan C, Springer, 2011
3. Motor Graders, Roninson E. G., Mir Publications, 1985
4. Tractors and Automobiles, Rodhiev and Rodhiev, MIR Publishers, 1984

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AE412) AUTOMOTIVE INSTRUMENTATION AND TESTING

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Automotive Chassis, Automotive engines and Automotive Electronics

#### COURSE OBJECTIVES:

- To learn basics and working principles of automotive sensors
- To identify and troubleshoot the problems in instrument cluster
- To understand the calibration of vehicle servicing instruments
- To identify the suspension and stability for directional control for vehicle
- To provide test procedures for wheels and braking performance

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Present basics of automotive sensors

**CO-2:** Explain the fundamentals of electronics and on-board diagnostics

**CO-3:** Discuss vehicle servicing instruments

**CO-4:** Review the suspension and stability for directional control for vehicle

**CO-5:** Assess the wheel testing and braking performance

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 3    | -    | -    | 3    | -    | -    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-2 | 3                     | 3    | -    | -    | 3    | -    | -    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-3 | 3                     | 3    | -    | -    | 3    | -    | -    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-4 | 3                     | 3    | -    | -    | 3    | -    | -    | -    | 2    | 2     | -     | 2     | 3                               | 3     |
| CO-5 | 3                     | 3    | -    | -    | 3    | -    | -    | -    | 2    | 2     | -     | 2     | 3                               | 3     |

#### UNIT-I:

**Introduction to Transducers and Sensors:** Transducers – types – resistive, capacitive and inductive based sensors with linear transfer characteristics, thermistor, LVDT, inductive pickup capacitance, strain gauges, semiconductors, piezoelectric accelerometer, proximity sensors, micro switches, encoders, piezoelectric pressure sensors, instruments – ammeter, voltmeter, speedometer, pressure gauge, vacuum gauge, analog and digital calibration and cathode ray oscilloscope.

#### UNIT-II:

**Vehicle Instrument Cluster:** Typical INS cluster, analog and digital dash instruments, speedometer, odometer, temperature, pressure, ABS, signalling circuits, seat belt restrainer, fuel level, tyre pressure monitoring, infotainment and telematics, diagnostic trouble codes and on-board diagnostics.

**UNIT-III:**

**Vehicle Servicing Instrumentation:** Wheel alignment gauges, laser alignment, exhaust gas analyzer, flasher instrumentations, wheel balancing, calibrations, dynamometer, starter motor, dynamometer calibrations, fuel ignition calibration, ignition timer calibration, stroboscope, tachometer generator, tyre air pressure instrument, head light alignment, head light intensity study, smoke meter, macro inspection of interior parts using fibre optics and boroscopes.

**UNIT-IV:**

**Suspension and Stability for Directional Control:** Measurement of dimensional and geometric characteristics, measurement of centre of gravity position, measurement of moments of inertia, measurement of suspension kinematic characteristics, measurement of suspension elastic, coulomb friction characteristics and measurement of shock absorber characteristics.

**UNIT-V:**

**Wheels and Braking Performance Test:** Dynamic cornering fatigue, dynamic radial fatigue tests – procedure, bending moment and radial load calculations. Impact test – road hazard impact test for wheel and tyre assemblies, test procedures, failure criteria and performance criteria. Bumpers - types of tests, pendulum test, fixed collision barrier test, procedure, performance criteria. Air and hydraulic brake test, air brake actuator, valves test, performance requirements. Parking brake – drawbar pull test and grade holding test

**TEXT BOOKS:**

1. Diagnosis and Troubleshooting of Automotive Electrical, Electronics and Computer Systems, Halderman J., Professional Technical Series, Pearson Education, 2012
2. Measurement Systems – Application and Design, Ernest O. Doebelin, McGraw-Hill, 2010

**REFERENCES:**

1. Automotive Mechanics, Crouse W. H., Anglin D. L., Tata McGraw-Hill, 2004
2. Instrumentation, Rangan, Mani and Sharma, Tata McGraw-Hill, 2004
3. Mechanical Measurements, Beakwith T. G., Buck N. L., Pearson Education, 2007
4. Advanced Automotive Fault Diagnosis, Tom Denton, Elsevier Butterworth-Heinemann, 2011

# VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

## B.Tech. VIII Semester

### (22PE1AE413) ROBOTICS IN AUTOMOTIVE INDUSTRY

| TEACHING SCHEME |     |   |
|-----------------|-----|---|
| L               | T/P | C |
| 3               | 0   | 3 |

| EVALUATION SCHEME |    |     |     |       |
|-------------------|----|-----|-----|-------|
| SE                | CA | ELA | SEE | TOTAL |
| 30                | 5  | 5   | 60  | 100   |

**COURSE PRE-REQUISITES:** Fundamentals of Electrical and Electronics Engineering and Theory of Machines and Mechanisms

#### COURSE OBJECTIVES:

- To provide a comprehensive understanding of robotics
- To study automation systems in manufacturing, focusing on automation levels, programmable logic controllers (PLCs), sensors, and actuators
- To learn robot programming languages, path planning algorithms, and motion control strategies
- To explore advanced robotics technologies, including artificial intelligence (AI), collaborative robots, IoT and digital twins

**COURSE OUTCOMES:** After completion of the course, the student should be able to

**CO-1:** Explain the fundamental principles of robotics, kinematics, and dynamics of robotic systems

**CO-2:** Discuss different levels of automation and use of PLCs in industrial automation

**CO-3:** Implement robot programming techniques and motion planning algorithms for various applications

**CO-4:** Appreciate the role of AI, collaborative robots, and emerging technologies in modern automation systems

**CO-5:** Apply robotic solutions to real-world industrial scenarios such as automotive manufacturing, autonomous vehicles, and sustainable automation practices

#### COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

| CO   | PROGRAM OUTCOMES (PO) |      |      |      |      |      |      |      |      |       |       |       | PROGRAM SPECIFIC OUTCOMES (PSO) |       |
|------|-----------------------|------|------|------|------|------|------|------|------|-------|-------|-------|---------------------------------|-------|
|      | PO-1                  | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 | PO-10 | PO-11 | PO-12 | PSO-1                           | PSO-2 |
| CO-1 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-2 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-3 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-4 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |
| CO-5 | 3                     | 1    | 1    | 1    | -    | 2    | 2    | -    | 2    | 2     | -     | 2     | 2                               | 3     |

#### UNIT-I:

**Fundamentals of Robotics:** Introduction to robotics – definition, history, evolution, components of robots, classification, overview of forward and inverse kinematics and dynamics for robotics applications, end effectors and grippers – types, selection criteria and applications.

**UNIT-II:**

**Automation Systems in Manufacturing:** Automation levels and their significance, benefits and challenges of automation. Programmable logic controllers – basics, ladder diagrams, role of PLCs in automotive manufacturing, sensors and actuators.

**UNIT-III:**

**Robot Programming and Path Planning:** Introduction to robot programming languages, offline vs online programming, motion planning algorithms – trajectory generation, obstacle avoidance, concepts of workspace and configuration space.

**UNIT-IV:**

**Advanced Robotics and Automation:** Applications of artificial intelligence techniques in robotics and Automation. Collaborative robots – concept and design, safety measures. Emerging technologies – IoT in robotics and automation and digital twins.

**UNIT-V:**

**Case Studies and Applications:** Robotics in automotive production lines – assembly, material handling, inspection and quality control. Automation in autonomous vehicles – role of robotics and sensors in ADAS and self-driving cars, path planning. Sustainability – role of robotics in green manufacturing and recycling and energy efficient automation.

**TEXT BOOKS:**

1. Robotics: Control, Sensing, Vision, and Intelligence, K. S. Fu, R. C. Gonzalez, and C. S. G. Lee, McGraw-Hill Education, 2017
2. Automation, Production Systems, and Computer-Integrated Manufacturing, Mikell Groover, 4<sup>th</sup> Edition, Pearson Education, 2014

**REFERENCES:**

1. Intelligent Vehicle Technologies: Theory and Applications, Vlacic, Michel Parent and Fumio Harashima, Butterworth-Heinemann Publications, Oxford, 2001
2. Autonomous Vehicle Technology-A Guide for Policymakers, James M. Anderson, Nidhi Kalra, Karlyn D. Stanley, Paul Sorensen, Constantine Samaras, Oluwatobi A. Oluwatola, RAND Corporation, 2016
3. Autonomous Driving - Technical, Legal and Social Aspects, J. Christian Gerdes, Barbara Lenz, Hermann Winner, Springer, 2016